

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

IV Semester of MSc (MI) Examination May 2018

Course code & Name: MI 852 Quality Control and Assurance

Date: 3/5/2018 Day: Thursday Time: 10:00am To 10:30 am

Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q – I Choose the correct answer for the following questions.

20

- 1) The following is usually carried out by a quality control personnel
(i) sampling (ii) specifications (iii) sale and marketing (iv) all of the above
- 2) Total Quality Management emphasizes most on the satisfaction of the needs and expectations of the following
(i) community (ii) employees (iii) leadership (iv) FDA
- 3) Preparations containing live microorganisms may be made or containers filled in areas having the following facility(ies)
(i) sterile air under positive pressure (ii) devices for dry heat sterilization
(iii) fumigation gases (iv) all of the above
- 4) A intravenous preparations should be free of the following validated by test for endotoxins
(i) pyrogens (ii) microorganisms (iii) chemical contaminant (iv) viruses
- 5) Though the quality is the responsibility of all in the industry, the major responsibility lies with the following:
(i) production department (ii) QC Manager (iii) CEO (iv) purchase department
- 6) The following products should be tested for antimicrobial effectiveness test
(i) eye drops (ii) injectable antibiotics (iii) transfusion fluids (iv) tablets

- 7) The following include all original records and documentation, which are the results of original observations and activities in a study/test
 - (i) test certificate (ii) test report (iii) specification (iv) raw data
- 8) An effective quarantine minimizes the chances of introduction of the following into an established colony of laboratory animals in animal house
 - (i) pathogen (ii) stability (iii) good hygiene (iv) all of the above
- 9) If the counting method used in pharmaceutical testing fails suitability test in the presence of product, the following should be done
 - (i) retest using same technique (ii) drop the test
 - (iv) retest using modified technique (iv) report as fail
- 10) Bacterial inoculums may be stored under refrigeration (2-8° C) for up to the following after preparation
 - (i) 1 day (ii) 2 days (iii) 7 days (iv) 2 hours
- 11) Media volume may be increased for performing the sterility test if the sample contains the following
 - (i) antimicrobial activity (ii) surfactants (iii) preservatives (iv) all of the above
- 12) Rinse fluid A supplemented Polysorbate 80, adjust to pH 7.1 ± 0.2 is required for Sterility test of products containing the following
 - (i) antibiotics (ii) antimicrobial agent (iii) oil (iv) surfactant
- 13) Experiments involving indigenous or exotic agents with potential aerosol transmission; may have serious or lethal consequences may be tested under the following facility
 - (i) BSL I (ii) BSL II (iii) BSL III (iv) BSL IV
- 14) A radionuclide in a kit of radiopharmaceutical preparation may be diluted in the following solution
 - (i) non-radioactive (ii) isotopic (iii) non-isotopic (iv) radioactive
- 15) Self-inspection is constituted to take the following measures for implementation of cGMP:
 - (i) penalize technician (ii) detect short comings
 - (iii) reject raw material (iv) all of the above
- 16) A following may comprise one or more lot identified packages or containers and may include material belonging to more than one lot-identified batch
 - (i) batch (ii) lot (iii) available sample (iv) consignment
- 17) Friability Test is generally carried out for the following pharmaceutical products
 - (i) ointment (ii) transdermal patches (iii) tablets (iv) cream
- 18) The following term describes a document that defines a set of step-by-step instructions to be followed for sampling, testing etc
 - (i) SOP (ii) protocol (iii) analytical technique (iv) specification
- 19) The following product functionality is related to the basic physical and chemical properties, such as composition and components distribution, surface properties or morphology of a product
 - (i) fundamental (ii) functional (iii) technical (iv) sensory
- 20) The following does not indicate a benchmark of quality
 - (i) ISO certification (ii) Underwriters Lab (UL) (iii) ICH (iv) all of the above

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

II Semester of MSc (BT/MI) Examination October 2017

Course code & Name: MI 852 Quality Control and Assurance

Date: 3/5/2018 Day: Thursday Time: 10:30am To 1:30 pm Maximum Marks: 50

Instructions:

1. Section I and II must be attempted in TWO ANSWER SHEET.
 2. Make suitable assumptions and draw neat figures wherever required.
 3. Use of non-programmable calculator is allowed.
 4. Show necessary calculations.
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	SECTION -I	20
Q – II A	Write a short note on any two of the following:	(06)
(i)	Product functionality	
(ii)	Laboratory facilities for GLP	
(iii)	ISO series with reference to quality of a product	
Q – II B	Discuss briefly: Concepts of TQM	(04)
	OR	
Q – II B	Discuss briefly: CPCSEA guidelines for animal house facility	(04)
Q – II C	Attempt any two of the following:	(06)
(i)	Discuss briefly: Protocol to Conduct a clinical laboratory study	
(ii)	Differentiate between: Quality control and quality assurance in pharmaceutical industries	
(iii)	List the key SOPs required in a testing facility. Discuss SOP for handling OOS results in brief	
Q – II D	Discuss briefly: IPQC and FPQC tests for pharmaceutical tablets	(04)
	OR	
Q – II D	Discuss briefly: Sampling process and sampling facilities in a pharmaceutical industry	(04)
	SECTION - II	30
Q – III A	Write a short note on any two of the following:	(06)
(i)	Components of kit for radiopharmaceutical preparation	
(ii)	Bioburden estimation of medical devices	
(iii)	Importance of sterility testing of pharmaceutical products	
Q – III B	Differentiate between any four of the following:	(04)
(i)	GMP and cGMP	
(ii)	Quality control and Quality by design	
(iii)	Biosafety level and risk group	
(iv)	Bioburden and antimicrobial effectiveness tests	
(v)	Bioassay and bioburden	
Q – III C	What is the importance of Antimicrobial effectiveness test? Discuss briefly: Suitability of counting method in the presence of product during the same.	(04)
	OR	
Q – III C	What is the importance of keeping controls during sterility testing of a product? Discuss briefly the system control in sterility testing of a pharmaceutical product.	(04)

Q – III C Storage of samples in a quality control laboratory is very important. **(01)**

Q – IVA Why should intravenous injectables be free of pyrogens? Discuss the LAL test in detail. **(04)**

OR

Q – IVA **Discuss briefly:** Test organism, inoculum preparation and standardization required for Antibiotic potency test of a drug. **(04)**

Q – IVB **Attempt any two of the following:** **(06)**

(i) **Discuss briefly:** Clean Room Activities during sterility testing of a pharmaceutical product

(ii) Give the outlines of the test procedure for microbial examination of non-sterile products

(iii) The importance of cGMP in a pharmaceutical industry

Q – IVC Discuss briefly: The Standard Microbiological Practices for various Biosafety Levels in biomedical and microbiology Laboratory **(03)**

OR

Q – IVC Discuss briefly the basis for the Classification of Biohazardous Agents by Risk Group **(03)**

Q – IVD **Justify:** Appropriate positive and negative controls help in interpretation of results **(02)**